

Concurrency → 4 classes

Agenda :-

Processes & Threads

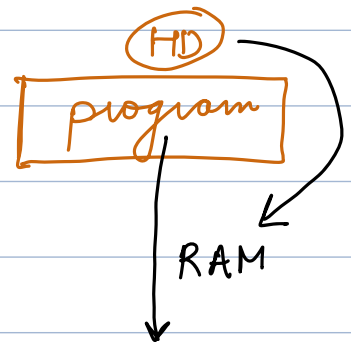
concurrency vs parallelism

Multitasking

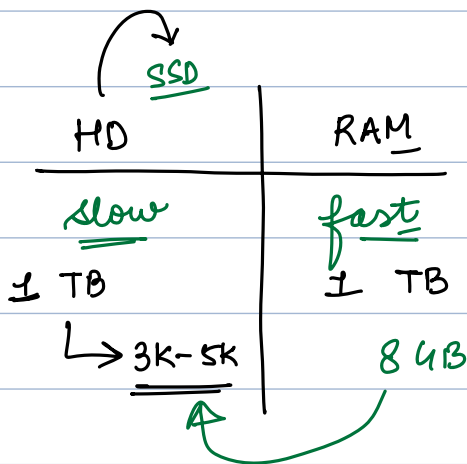
Threaded program in Java

• apk • dmg • exe

code $\xrightarrow{\text{compile}}$ executable file $\xrightarrow{\text{download \& install}}$



program : process
in execution

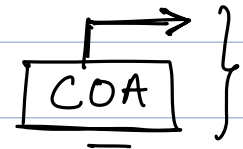


CPU $\xrightarrow{\times}$ HD

→ RAM

2.2 GHz

↓
{ 2.2×10^9 ins/sec



Process :- program in execution



PCB (Process Control Block)

data structure which stores info about a process.

class ProcessControlBlock {

pid ;

list < variable >

registers

priority

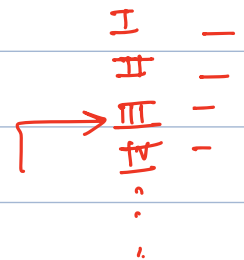
memory details

program counter

call stack

⋮

}



current line of execution.

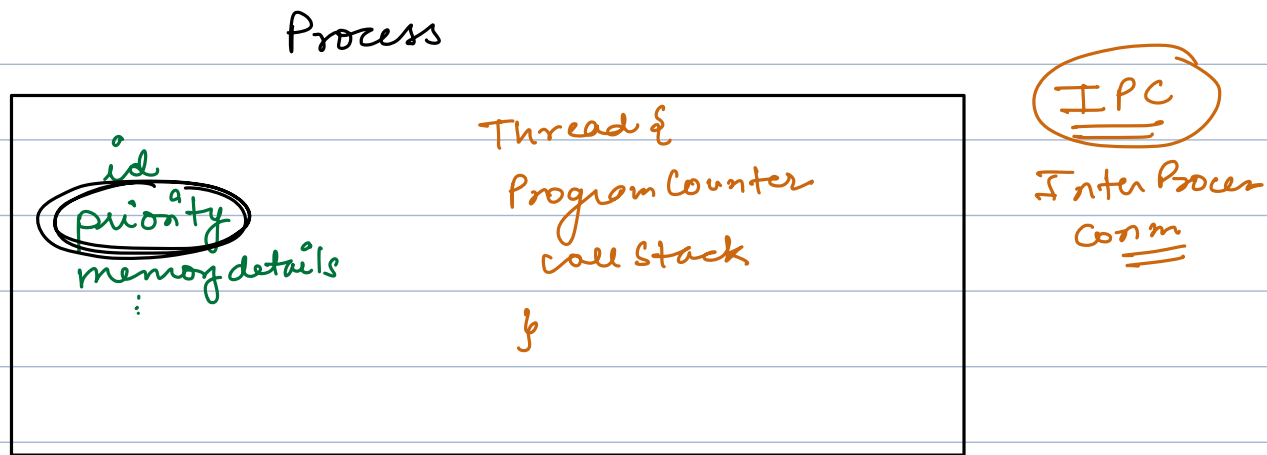
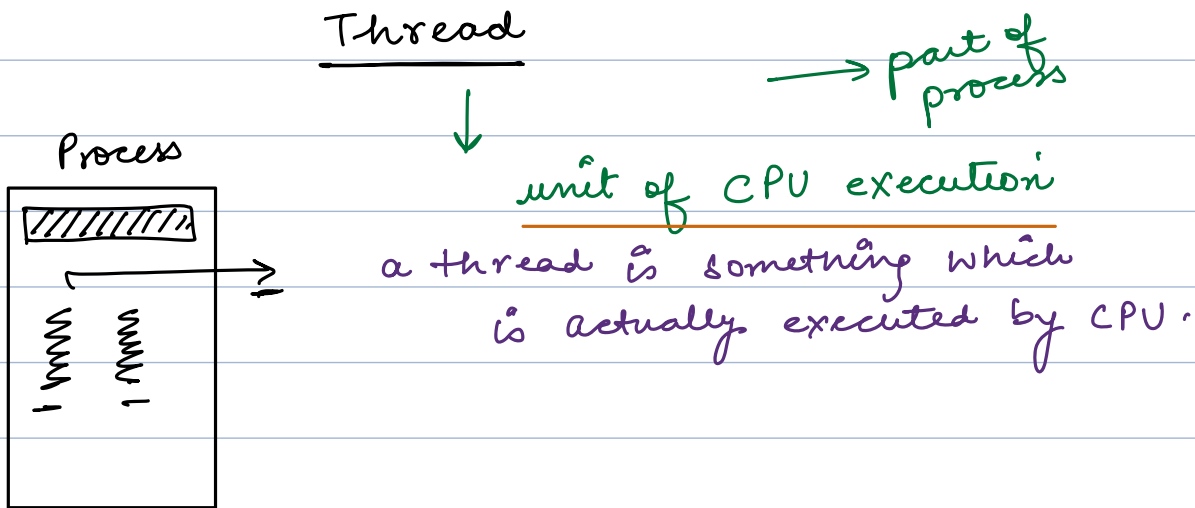
MSWord

consuming input — P1

displaying the input — P2

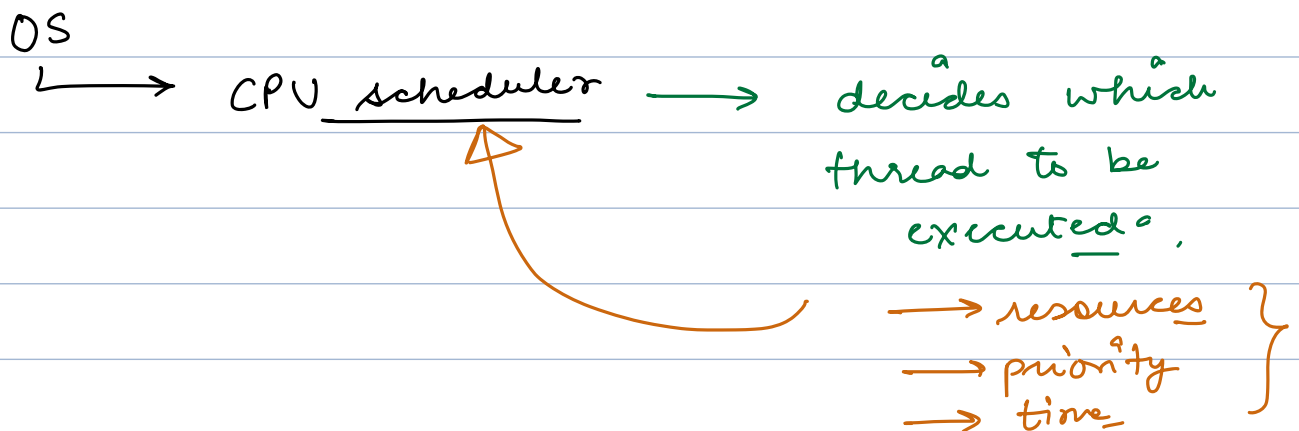
autosaving — P3

auto correct / auto suggestion — P4



① Data sharing is much easier in threads.

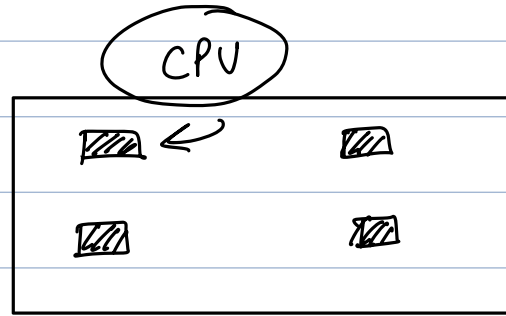
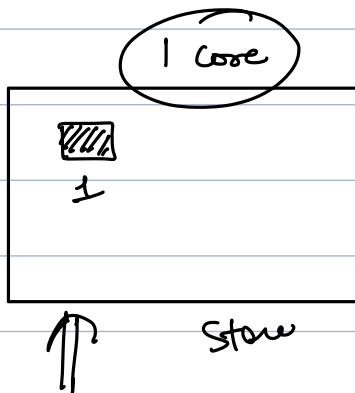
② creation of threads has less overhead.



CPU



single core vs Multicore



Each core can execute one thread at a moment o.

i3 - dual

i5 - quad

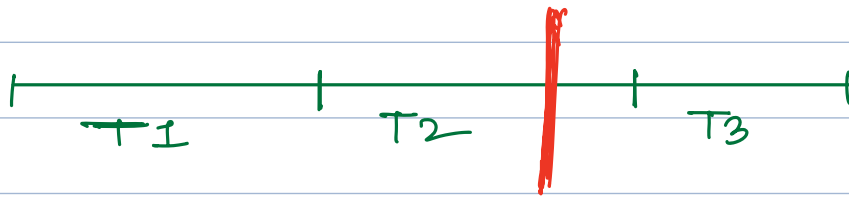
i7 - quad + Hyperthreading

2 threads

concurrency vs parallelism

Case I
single core

→ until one thread completes the execution, the other thread can't start



① How many threads are partially completed?

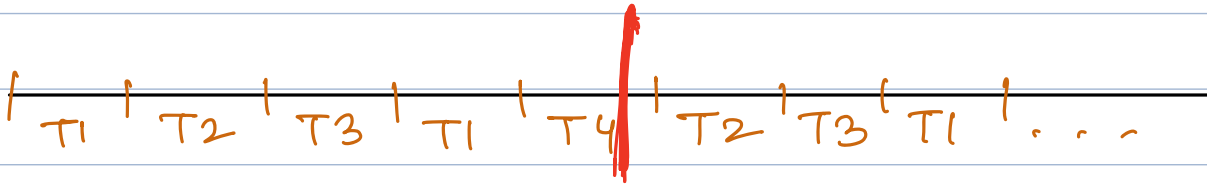
①

② How many threads are making progress

↓
①

Case II

Single core
switching b/w
threads is allowed



① How many threads are partially completed?

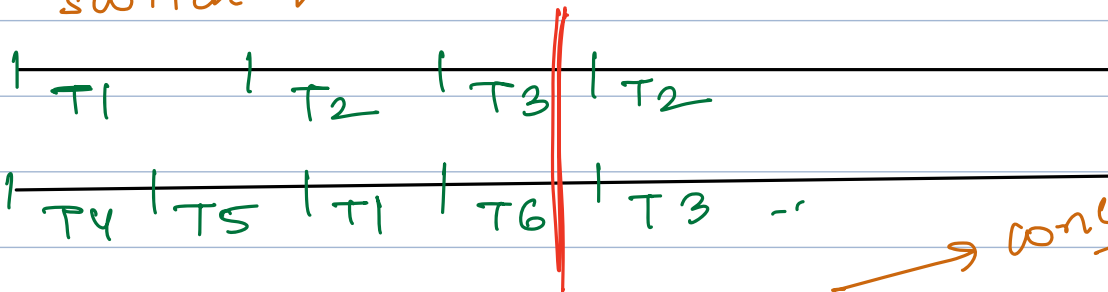
≥ 1

② How many threads are making progress

$= 1$

Case III

multi-cores
switch ✓



① How many threads are partially completed?

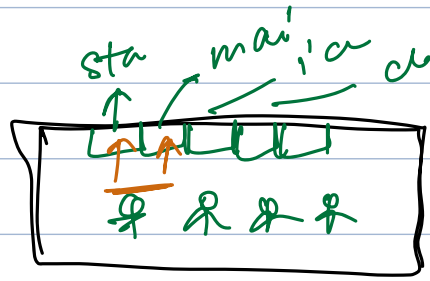
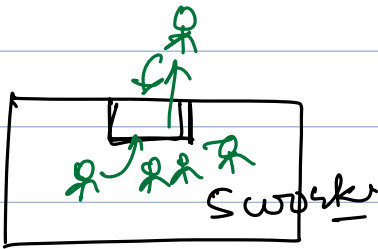
many

→ concurrency

② How many threads are making progress

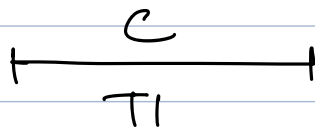
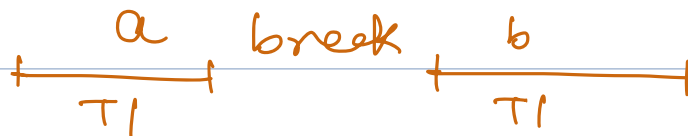
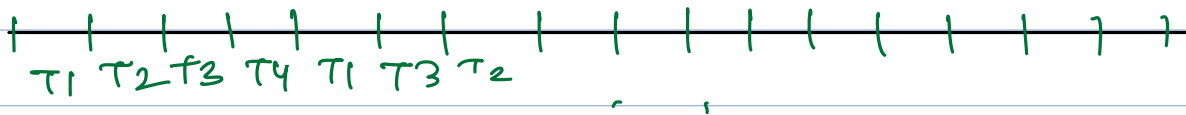
> 1

→ parallelism



Stores

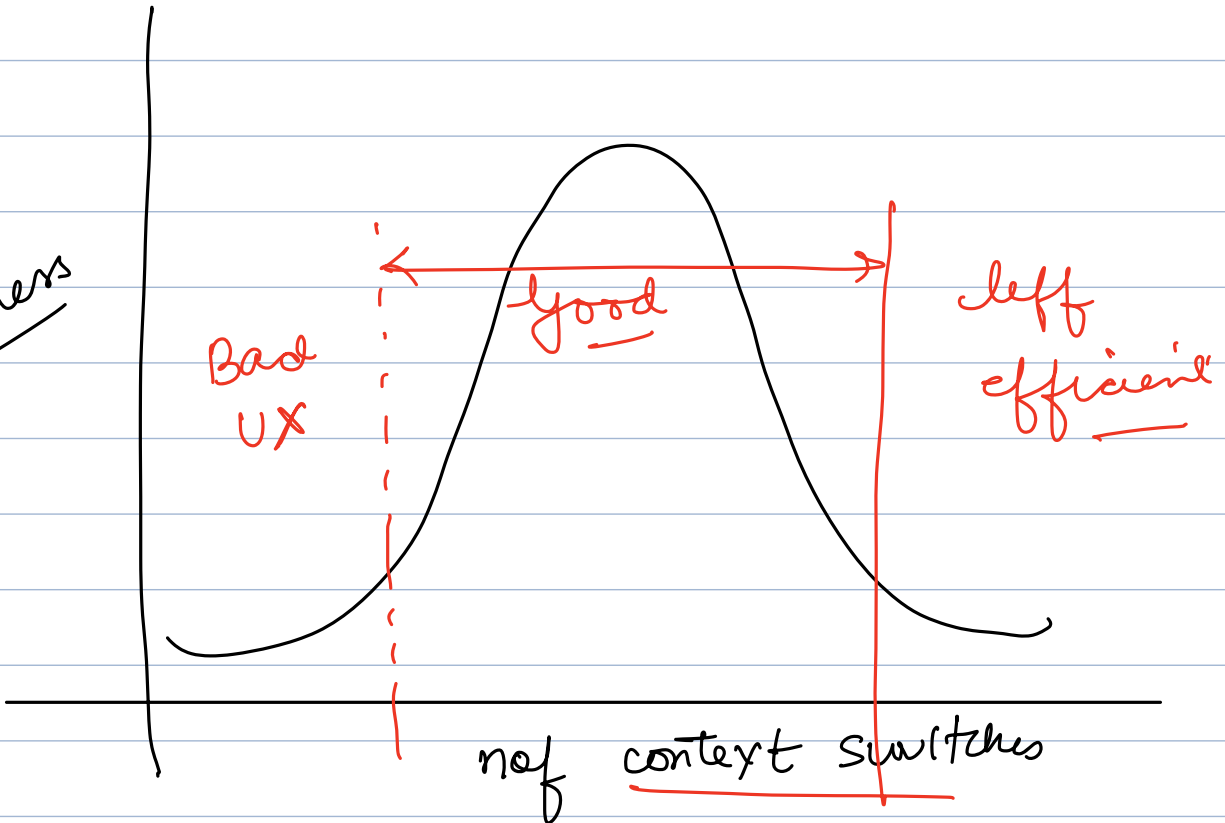
context switch



- 1) $a + b = c$
- 2) $a + b < c$
- 3) $a + b \geq c$

$T1 \rightarrow T2$

- ① save the curr state of $T1$
 - ② load the pr state of $T2$
- Time ✓



break: 10:26pm

Java: Threaded program

① define the task.



create a class for the task.

```
class HelloWorldPrinter {
```

```
    ==
```

```
}
```

② implement Runnable interface.



```
run()
```

```
==
```



code that
perform the task.

// Now you can't extend any more class

// Composition vs Inheritance

// In this you always have to write a new thread but
the runnable way can use lambda expressions

// Effectively thread also implements Runnable , so
we are not actually overriding the method of
Thread class